

Power dynamics

Adding complexity to portable electronic equipment usually means increased power consumption, but despair not. Here's more power for those hungry handhelds

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One of the smaller revolutions within the digital age has been the proliferation of portable electronics that help people on the go stay in touch, keep themselves organised and get on with their work. However, the current life of batteries impedes the spread of promising new technologies and features. For instance, 3G (third generation wireless technology) allows for full motion video, audio downloading and high-speed data, but at the cost of ridiculously low battery life (less than an hour in trial conditions). Thus, mobile devices have to walk the tightrope between providing new whiz-bang features and economising on power at the same time. Internet, convergence, connectivity, higher processing power, and new wireless technologies may very well push current battery technology into obsolescence.

R&D teams at various corporations and alternative energy researchers are fiercely competing with each other to capture the \$6 billion battery market. Researchers are discovering new ways to maximise battery life, to make devices more independent, perhaps self-sustaining. Battery technologies are entering new frontiers with exotic cocktails based on modern polymers,

metals and carbides. Researchers are also thinking out of the box, tapping into solar and kinetic energy. It might not be long before you are wearing polymer boots and solar vests that recharge your MP3 player, your cell phone and your notebook.

Lithe Lithium

Most laptops and mobile phones today work on Lithium-ion batteries. They are lightweight and rechargeable, albeit expensive and toxic. But as of now, Lithium-ion batteries have reached their physical limit in terms of storage power. Of course, new developments currently taking place could improve performance drastically. Matsushita Industrial, better known as Panasonic, has developed a coin-type lithium-ion battery capable of 10 times as much current as existing cells. The new battery has a capacity of 140 milli-Ampere hours (mAh) and can be recharged up to 500 times. The extra voltage could power next generation portable devices such as GPS devices and terminals. Scientists at Brookhaven National Laboratory are trying out a cheaper and less toxic Boron-based additive. This development opens up new possibilities for electrolytes that can be developed for the Lithium-ion batteries. If the development cycle works out as planned, the new batteries may be on the market in two to three years.

Imaging: Neeta Wadiker